

Longitudinal Data Analysis

LMM, GLMM, GEE

——数据之间有correlation: heterogeneity就是within subject的correlation

1. Questions of interest:
 - Does the distance change over time?
 - What is the pattern of change?
 - Is the pattern of change different for boys and girls? How?
2. 先画图! 看数据趋势&outliers
3. 注意哪些variables是time-varying的哪些是time-invariant的

Mixed-effects models

Linear mixed-effects models (LMM)

naive models: 一个线性模型拟合所有data, 没有考虑到correlation; one model for one subject 无法model共同特征, 无法做预测, 且parameter太多了

LMM: population level+random effects(index by i)+random error

$$Y_{ij} = X_{ij}^T \beta + Z_{ij}^T b_i + \epsilon_{ij}$$

i : subject, j : observation

- β : fixed effects
systematic part of the response
- b_i : random effect $Z_{ij} \subset X_{ij}$
通常就取截距 intercept (有时候也会加上随时间的斜率 slope, 如果数据并不parallel)
反映人和人的异质性
必须假设服从于均值为0方差任意的正态分布——也体现出其局限性。最后我们估计的parameter其实是这个分布的cov matrix
类似与population mean的deviation
- ϵ_{ij} random error $N(0, \sigma^2)$, 与 random effect 独立

1. 写成矩阵形式有 $Y_i = X_i \beta + Z_i b_i + \epsilon_i$

$$\begin{aligned}\text{subject-specific : } E(Y_i | b_i) &= X_i \beta + Z_i b_i \\ \text{marginal : } E(Y_i) &= X_i \beta\end{aligned}$$

2. $Var(Y_i) = Z_i D Z_i^T + \sigma^2 I_{n_i} \equiv \Sigma_i(\theta)$

例如当bi只包含random intercept时, 有

$$\text{Var}(\mathbf{Y}_i) = \begin{bmatrix} d + \sigma^2 & \cdots & d \\ \vdots & \ddots & \vdots \\ d & \cdots & d + \sigma^2 \end{bmatrix}$$

两项方差的建模不要太fancy，否则会遇到identifiability的问题！

3. pooling all data together

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{b} + \boldsymbol{\epsilon}$$

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}, \quad \mathbf{Z} = \begin{pmatrix} Z_1 & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \cdots & Z_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}, \quad \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}$$

$$\mathbf{Y} \sim N(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma}(\boldsymbol{\theta})), \text{ where}$$

$$\boldsymbol{\Sigma}(\boldsymbol{\theta}) = \begin{pmatrix} \boldsymbol{\Sigma}_1(\boldsymbol{\theta}) & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \cdots & \boldsymbol{\Sigma}_n(\boldsymbol{\theta}) \end{pmatrix}$$

- 这时我们就可以写likelihood.
- 最终目的是估计 β (the parameter) , random effect 只是用来解释correlation structure

4. LMM的特点:

- Extremely flexible in handling imbalance: 可以有缺失时间、时间点不对齐
- Separation of variance into between-subject ($\mathbf{b}_i$) and within-subject (ϵ_{ij}) components, allowing parsimonious modeling for covariance among repeated measures
- Prediction of individual response trajectories: 这个GEE没有

5. β interpretation

- subject-specific: beta is the effect of X on the subject-specific mean response
- population-average

LMM都可以!

remark: 这在GEE中不成立，其每个人的average后会变得很小(每个人都是binary, logistic, average后斜率就会很小。when model is nonlinear, there is no coincidence!

6. 分层嵌套(nested)的grouping factor: 如何建立

- 不要过分复杂!

7. 因为是fully parametric的所以可以MLE

- log likelihood 迭代, 先fix θ 可以得到 β 的闭式解, 然后可以再解 θ , 迭代直到收敛。
 - 经典方法为EM algorithm
- REML (Restricted Maximum Likelihood Estimator, also called residual maximum likelihood)

两种思路:

- projection
- integrate out β

8. prediction ... of random effects: estimate an individual's trajectory

$$\text{BLUP: } \hat{b}_i = E(b_i | Y_i, \hat{\beta}, D, \sigma^2) = DZ_i^T \Sigma_i^{-1} (Y_i - X_i \hat{\beta})$$

代入即得individual-specific mean trajectory

$$\hat{Y}_i = X_i \hat{\beta} + Z_i \hat{b}_i = X_i \hat{\beta} + Z_i DZ_i^T \Sigma_i^{-1} (Y_i - X_i \hat{\beta})$$

GLMM

1. Random component

$Y_{ij} (i = 1, \dots, n, j = 1, \dots, n_i) \sim \text{an exponential family distribution } f_{\theta_{ij}}(y)$

2. Systematic component

$$g(\mu_{ij}) = \eta_{ij},$$

- Mean: $\mu_{ij} = E(Y_{ij} | \mathbf{b}_i) = g^{-1}(X_{ij}^T \boldsymbol{\beta} + Z_{ij}^T \mathbf{b}_i)$ 是 $\mathbf{b}_i$ 的random variable

remark:

- $\boldsymbol{\beta}$ 是subject-specific的, 是固定住该个体自身随机偏差后的效应
- attenuation: $E(Y_{ij}) \approx \frac{\exp(X_{ij}^T c \boldsymbol{\beta})}{1 + \exp(X_{ij}^T c \boldsymbol{\beta})}$, $c = (1 + 0.346d)^{-0.5}$ d增大c减小
- Linear predictor: $\eta_{ij} = \mathbf{X}_{ij}^T \boldsymbol{\beta} + \mathbf{Z}_{ij}^T \mathbf{b}_i$, $\mathbf{b}_i \sim N(\mathbf{0}, \mathbf{D})$
- Link function: g , a known, smooth monotonic function

3. marginal mean没有闭式解 (除非 $g=id$)

4. $var = \phi V$, $cov(Y_{ij}, Y_{ij'} | \mathbf{b}_i) = 0$

The marginal variance, covariance

$$\begin{aligned} \text{Var}(Y_{ij}) &= \text{Var}[E(Y_{ij} | \mathbf{b}_i)] + E[\text{Var}(Y_{ij} | \mathbf{b}_i)] \\ &= \text{Var}[g^{-1}(\mathbf{X}_{ij}^T \boldsymbol{\beta} + \mathbf{Z}_{ij}^T \mathbf{b}_i)] + E[\phi V(\mu_{ij})] \\ \text{Cov}(Y_{ij}, Y_{ij'}) &= \text{Cov}[E(Y_{ij} | \mathbf{b}_i), E(Y_{ij'} | \mathbf{b}_i)] + E[\text{Cov}(Y_{ij}, Y_{ij'} | \mathbf{b}_i)] \\ &= \text{Cov}[g^{-1}(\mathbf{X}_{ij}^T \boldsymbol{\beta} + \mathbf{Z}_{ij}^T \mathbf{b}_i), g^{-1}(\mathbf{X}_{ij'}^T \boldsymbol{\beta} + \mathbf{Z}_{ij'}^T \mathbf{b}_i)] \end{aligned}$$

5. 局限性:

- 认为residual之间independent, does not allow within-subject source of correlation (i.e., AR(1))
- base on likelihood, 需要对random effect做积分, 一般没有显式解

因此我们提出以下estimation方法:

- Gauss-hermite quadrature methods: 数值逼近。可以用AIC选模型。only allow the random intercept and "family=binomial/poisson"
- Penalized quasi-likelihood methods: analytical approximation of the likelihood (faster, but less accurate)。不能用AIC做模型选择。

综上其用的没有GEE多。

GEE

extend of quasi likelihood。关注其和GLMM的优劣以及他们不同的对参数的interpretation

mean + (variance + **working correlation model**)

mean与之前的一样, 要求必须正确。每个人认为是一样的。

var变成了矩阵, 考虑人的关系。

1. working correlation matrix $R_i(\alpha)$

- $R_i(\alpha)$ 可以因为观测数量不同导致每个个体维度不同但是生成机制必须一样 (只有一个 α)
- 叫working... 统计学的惯例, 指的是其可以是misspecified
- 如何选择working matrix:

- independent→identity: 大家发现好像挺好的.....误设了也consistent
- unstructured: Require balanced data (每个人观测的时间点相同) 一般是做benchmark
- exchangeable (compound symmetric) : 任两个时间点之间corr=α. 常用
- m-dependent (banded Toeplitz) $Corr = \alpha_{|j-k|}$, for $|j-k| \leq m$
此时要求time是equally spaced。常用1-dependent
- autoregressive of order 1 (AR1): 要求time是equally spaced
- Markov: 不要求时间间隔相同 $corr = \alpha^{|t_j-t_k|}$
 - 注意根据组间还是组内更重要来选择working matrix, 例如生物统计中就常用exchangeable因为关注的是组间差异

2. $\sqrt{n}$ -consistent 这里的n指的是样本量!

- $\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} N(0, A^{-1}BA^{-1})$
- $A = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n D_i^T V_i^{-1} D_i$
- $B = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n D_i^T V_i^{-1} (Y_i - \mu_i)(Y_i - \mu_i)^T V_i^{-1} D_i$

Conditional logistic regression

- 很像GLMM (match pair data 也可以看成logitudinal的一个简单版本) , McNemar
- 1. subject-specific model: $logit(P(Y_{it} = 1)) = \alpha_i + \beta X_{it}$: 每个人需要单独的 α_i , 这体现了人的异质性。但是与GLMM假设intercept有分布不同的时, 这里视为nuisance parameter之后删掉。

(for subject i , the odds of $Y_{i2} = 1$ are $\exp(\beta)$ times the odds of $Y_{i1} = 1$)

2. conditional likelihood $CL(\beta) = \prod_{i:s_i=1} \left(\frac{1}{1+\exp(\beta)} \right)^{Y_{i1}} \left(\frac{\exp(\beta)}{1+\exp(\beta)} \right)^{Y_{i2}} = \frac{\exp(\beta n_{12})}{[1+\exp(\beta)]^{n_{12}+n_{21}}}$

另一个视角: 认为数据是独立的, 即定义新的响应变量则可以用普通的logistic回归估计 β

$$Y_i^* = \begin{cases} 1, & (Y_{i1}, Y_{i2}) = (0, 1) \\ 0, & (Y_{i1}, Y_{i2}) = (1, 0) \end{cases}$$

3. MLE

$$\hat{\beta} = \log \left(\frac{n_{12}}{n_{21}} \right), \quad SE = \sqrt{\frac{1}{n_{21}} + \frac{1}{n_{12}}}$$

4. matched case-control data: Y是matched (一个0一个1, 即一个得病一个不得病) , 要看X(exposure)

- OR亦适用
- interpret

• For the MI data, $\exp(\hat{\beta}) = 37/16 = 2.31$

• Interpretation 1: for each case-control pair, the odds of being diagnosed as having diabetes in MI cases are 2.31 times the odds in MI controls **study design**

• Interpretation 2: for each case-control pair, the odds of having MI in diabetic patients are 2.31 times the odds in non-diabetic people **事情发生顺序**

Survival Analysis

introduction

1. 与

- event death, disease occurrence/recovery...
 - survival time (time-to-event)
 - censoring, 删失信息的处理。这是生存分析最大的特点，后续我们考虑的就是如何尽可能好地处理删失
- 在后续建模中通常假设是non-informative（因为搬家等原因自行退出）的，否则仍然需要对其建模。如何处理每个删失的样本是重要的
- 大家主要研究右删失，其中常用independent censoring assumption: censoring time & event time are independent (也可以conditioning on covariates)——这意味着留下的信息是无偏的
- survival time 是结局变量

2. notation

Ti time to event, Ci censoring time
Yi=min{Ti,Ci} Survtime
 δ_i status 1=event, 0=censored
(Ti,Ci) underlying survival data; (Yi, δ_i) observed

patient	Survtime	Status
1	7	0
2	6	1
3	6	0
4	5	0
5	2	1
6	4	1

其中1表示event，0表示删失。

3. 生存函数 $S(t) = P(R > t) = 1 - F(t)$ 解释：subject survive past time t的概率。单增， $S(0)=1, S(\infty)=0$ 但是实际上不能直接1-F* 因为删失的生存时间比观测的长。
- 风险函数 hazard function：生存过一个节点，那一下马上会死亡的概率

$$h(t) = \lim_{\delta \rightarrow 0^+} \frac{\Pr(t < T \leq t + \delta \mid T > t)}{\delta} = \frac{f(t)}{S(t)}$$

累计风险函数: 对上式积分

$$h(t) = -\frac{\partial \log(S(t))}{\partial t}; \quad H(t) = -\log(S(t)); \quad S(t) = \exp(-H(t))$$

- Mean survival time $\mu = E(T) = \int_0^\infty t f(t) dt = \int_0^\infty S(t) dt$

model

parametric

- 价值在于知道如何simulate
- 想要model一个东西, 先想range, 例如0到正无穷通常考虑gamma, 0到1通常考虑beta

1. 指数分布。hazard= λ

2. Weibull distribution

hazard: $h(t) = \alpha \lambda^\alpha t^{\alpha-1}$, $\alpha, \lambda > 0$ $\alpha=1$ 时退化为指数分布, $\alpha > 1$ 时h单增, < 1 时单减

chf $(\lambda t)^\alpha S(t) = \exp(-(\lambda t)^\alpha) \mu = \lambda^{-1} \Gamma(1 + \frac{1}{\alpha})$

3. gamma 不常用, 因为time趋于正无穷时hazard趋于常数 (不符合现实情况), 且hazard和survival没有显式形式

estimation

- 本质上就是做survival time的histogram但是这里有censoring情况!

nonparametric estimation: Kaplan-Meier

1.
$$\hat{S}(t) = \prod_{t_i \leq t} (1 - \hat{q}_i) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

- n_i : number of subjects **at risk** at time t_i (包括所有活到 t_i 时间的人数: 接下来一秒死亡/censor/继续活的都算)
 - 删失通过其参与方程中。
- d_i : number of events at time t_i
 - 只需要取event time, 不用看censoring!
 - d_i 一定是非0的
- 具体步骤: 先对event time排序, 计算每个 t_i 的 n_i 和 d_i , 然后计算, 每次乘到上一个 S_i 上就行。

2. CI: 用delta method (先取log比较好求方差)

- To give the confidence interval for $\hat{S}(t)$, we need to estimate its variance

- Linear type: first get the variance for $\log(\hat{S}(t))$ (Greenwood's formula)

$$\hat{\sigma}^2(t) \equiv \text{Var}(\log(\hat{S}(t))) = \sum_{t_i \leq t} \text{Var}(\log(1 - \hat{q}_i))$$

$$\text{(use the delta method)} = \sum_{t_i \leq t} \left(\frac{\partial(\log(1 - \hat{q}_i))}{\partial \hat{q}_i} \right)^2 \text{Var}(\hat{q}_i) = \sum_{t_i \leq t} \frac{1}{(1 - \hat{q}_i)^2} \frac{\hat{q}_i(1 - \hat{q}_i)}{n_i} = \sum_{t_i \leq t} \frac{d_i}{n_i(n_i - d_i)}$$

Then use the delta method again to get

$$\text{Var}(\hat{S}(t)) \approx [\hat{S}(t)]^2 \text{Var}(\log(\hat{S}(t))) = [\hat{S}(t)]^2 \hat{\sigma}^2(t)$$

The confidence interval for $\hat{S}(t)$ is $(z_{1-\alpha/2}$ is the standard normal quantile)

$$\hat{S}(t) \pm z_{1-\alpha/2} \hat{S}(t) \hat{\sigma}(t)$$

Problem: may give confidence interval beyond 1 or below 0

21

常用log-log来限制范围, 就是在上面再log一下做delta method, 最终得到 $\hat{S}(t)^{\exp[\pm z_{1-\alpha/2} \log(\hat{S}(t)) \hat{\sigma}(t)]}$ **Shat应**
该有个inverse漏打了

Nelson-Aalen

先estimate H: $\hat{H}(t) = \sum_{t_i \leq t} \frac{d_i}{n_i}$

再 $\hat{S}(t) = \exp(-\hat{H}(t))$

- 在初期与上者渐进等价

```
surv_obj <- Surv(time = lung$time, event = lung$status == 2) # 先创建object且需要调整编码为event=1
```

```
fit_KM <- survfit(surv_obj ~ 1, conf.type= "log-log") # 默认为log
```

parametric estimation

- 关注joint likelihood的写法

$$L = \prod_{\delta_i=1} f(Y_i) \prod_{\delta_i=0} S(Y_i)$$
$$\log(L) = \sum_{\delta_i=1} \log(h(Y_i)) - \sum_{i=1}^n H(Y_i)$$

(notice f=h/S)

compare between

$H_0 : S_0 = S_1$ (or $h_0 = h_1$)

log-rank test

Log-Rank Test

- Log-rank test is the most popular method for comparing two survival functions
- $t_1 < t_2 < \dots < t_D$ are ordered, distinct times of event in either group
- At the i th time of event

	Control	Treatment	
Failure	d_{0i}	d_{1i}	d_i
Non-failure	$n_{0i} - d_{0i}$	$n_{1i} - d_{1i}$	$n_i - d_i$
	n_{0i}	n_{1i}	n_i

n_{ki} : number of subject at risk in group k at time t_i

d_{ki} : number of events in group k at time t_i

Log-Rank Test

- Under H_0 , d_{0i} given d_i, n_i, n_{0i} follows a hypergeometric distribution
 - Mean: $e_{0i} = E(d_{0i}) = \frac{n_{0i}d_i}{n_i}$
 - Variance: $v_{0i} = \text{Var}(d_{0i}) = \frac{n_{0i}n_{1i}d_i(n_i - d_i)}{n_i^2(n_i - 1)}$

- Summing up over all tables

$$U_0 = \sum_{i=1}^D (d_{0i} - e_{0i}), \quad V_0 = \text{Var}(U_0) = \sum_{i=1}^D v_{0i}$$

- Approximately, we have

$$\frac{U_0}{\sqrt{V_0}} \sim N(0, 1)$$

- This is the log-rank test

- 对于proportional hazards建模最powerful
- weighted:
 1. log-rank $w=1$
 2. Fleming-Harrington 重视开始的时候 (用 $\hat{S}(t_{i-1})^\rho$, 实际上就是取其前一点点的生存概率)
 3. Wilcoxon

log-rank for >2 groups

H0: K+1个S都一样

Assessing the effects of explanatory variables on survival time (Cox regression model)

1. Proportional Hazard(PH) assumption

$$h(t|X_i) = h_0(t) \exp(\beta_1 x_{i1} + \cdots + \beta_p x_{ip})$$

- interpretation: $\exp(\beta_j)$ =hazard ratio

2. AFT

$$\log(T) = X^T \beta + \varepsilon, \text{ 略。实际上还要考虑censoring}$$

- AFT: Multiplicative effects of predictors on survival time
- PH: Multiplicative effects of predictors on hazard
- weibull 分布的时候二者性质相似

3. Cox PH model: semiparametric: 不需要假设 $h_0(t)$ 的分布, 于是不能做MLE, 于是maximize **partial** likelihood 26五推问了生存分析怎么做, 希望的答案就是maximize **partial** likelihood

- 这个可以估计个体的。
- If there were only 2 groups and no ties in event times, then the score test and the log-rank test would be identical
- NPMLE: 写出likelihood, 认为是有一堆nuisance和我们关心的参数 β , 然后缩小到increasing step function的函数然后消掉nuisance, fix β 求出nuisance的MLE, 然后再带回来求 β 的mle

4. residuals

1. Cox-Snell Residual
2. martingale residual

5. influence statistic

extension

1. 分层Cox模型

- 如果有一个factor影响很大

2. time-dependent covariate model

杂录

MLE有什么性质: consistency, asymptotic normality, asymptotic efficiency

Multiple testing

motivation: under the null p is uniformly distributed. Thus if we try 1000 tests, we can't say one is significant with $p=0.03$. Here we need multiple testing.

- 高维统计里面得到 p value 第一步看 histogram

Estimate π_0 by a cutoff λ , e.g., $\lambda = 0.5$

$$\hat{\pi}_0 = \frac{\#\{p_i > \lambda; i = 1, \dots, m\}}{m(1 - \lambda)}$$

Statistical Genetics

基础知识

1. Single Nucleotide Polymorphisms (SNP)

genetic marker

2. 连锁不平衡 LD

$$r^2 = \frac{(p_{AB} - p_A p_B)^2}{p_A p_a p_B p_b}$$

(统计学上就是 pearson corr 的平方) 用于解释 p -value

存在区域聚集，与人种有关

3. genotype: AA/Aa/aa allele: A/a

Statistical Methods for Population-Based Association Analysis

study design

1. case-control: 基于疾病二分变量。患病的收，对照组如果和疾病组的人可以 match，就收。这样效率高但是有 bias 且不能估计患病率
2. 队列设计 Cohort Design
 - case-control 往往是其的 subset

test

1. Genotype test 2-df
2. Dominant/recessive 1-df
3. additive

GWAS